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Gamification from the Viewpoint of Motivational Theory

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Abstract

Gamification is defined as the use of game elements and game design techniques in non-game contexts. it is expected that global gamification market will grow to USD $5.5 Billion by 2018. This tremendous market growth relies on basic human instinct to get enticed by both extrinsic and intrinsic motivations. This study aims to provide insights into gamification approach from motivation perspective. For this purpose, conceptual foundations of gamification described, including the structure and Benefits of Gamification, and then related motivation theories reviewed.

Keywords:

Gamification;
Motivation;
Engagement;
Theories of Motivation.

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1- Introduction

Gamification is a strategy that employs game mechanics, techniques, and theory in areas that traditionally do not function like a game. The word can be traced back as early as 2004, but the concept goes back further. Although it is still a relatively new term, gamification has solid roots in organization motivation, customer loyalty, and business development. In fact, almost every one of us has experienced some form of gamification in our lives. For example, when's the last time you made a flight decision based on your frequent flyer status? Maybe you or your kids have taken some martial arts classes and leveled your way up with different colored belts. And the Boy Scouts have been motivating achievement through social recognition and rewards since 1910. But what is the base of gamification performance? Games it seems are perfectly tuned to give out rewards that engage the brain and keep us questing for more. Engagement occurs when the brain is rewarded, and that for something to be perceived as rewarding, it must evoke positive emotions in a person. Essentially, there are two components to the perception of something being rewarding: wanting and liking. Neuroscience researchers found that playing games releases high amounts of dopamine in the human brain. Dopamine is associated with increased learning, reinforcement of the current behavior, and attention. Furthermore, found that dopa-mine influences the incentive salience in general reward situations, i.e., the recipients want the experience more often. However, there is no mediation with hedonic impact, i.e., the recipients do not necessarily like the actual experience more. Gamification using the elements of the game in contexts that lack Gamification elements are trying to motivate people and engage them in various activities. The aim of this paper is to studying Gamification from the viewpoint of motivational theory.

2- Gamification

Deterding, Dixon, Khaled, and Nacke [1], provide a further definition of gamification as “the use of game design elements in non-game contexts.” In tracing the historical context of gamification and its place in the digital media industry its emergence dates back to 2008. The popularity effectively commenced around 2010 and has since 2010 managed to institutionalize itself as a common household term [1].

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Gamification is a *persuasive technology* that attempts to influence user behaviour by activating individual motives via game-design elements [2]. As a consequence, this approach does not deal with designing games that can generally be defined as solving rule based artificial conflicts or simulations [1, 3]. Thus, gamification needs to be contrasted to related concepts such as *serious games* and *games with a purpose*. Serious games reflect games that are linked to a particular learning objective [4].

Gamification encompasses the design of "gamified" service bundles [5]. These bundles comprise of a core offer (a product, a service or an information system) and an IT-based, gamified enhancing service for the core offer [6]. The starting point for designing gamified enhancing services are the specific usage objectives for the core offer as defined by core office providers. For instance, such usage objectives could be derived from the analysis of historic user behaviour and may describe how core offer providers wish their core offer to be used in future. These objectives are translated into appropriate game-design elements that are compiled into gamified enhancing services. By creating a game like usage experience, these enhancing services strive to activate individual user motives regarding the core offer [7] and consequently support its consumption. Moreover, bundling core offer and enhancing services also requires the adaption of the core offer to the enhancing service [8].

2-1- Potentials and Application of Gamification

The potential of gamification is based on comprehensive motivational support and on invoking flow experiences. Ryan and Deci [9] distinguish between intrinsic and extrinsic motivation. Intrinsic motivation is directly rooted in a given task, whereas extrinsic motivation always aims at external goals such as financial compensation. Traditional incentive mechanisms are usually based on in- creasing extrinsic motivation, i.e., introducing financial rewards. Such stimuli of- ten fail to increase motivation in the long run as adaptation effects undermine their effectiveness. However, IT-based gamified enhancing services are able to arouse the intrinsic motivation of users regarding a core offer [3]:

- Increase in user satisfaction: The continuous documentation of one's own behaviour visualizes progress, facilitates the derivation of achievable personal goals and offers immediate feedback so that users perceive feelings of high individual performance.
- Conveyance of optimism: Gamification enables self-determination as well as experiencing a sense of achievement, or more specifically the hope of experiencing success.
- Facilitation of social interaction: Gamification is usually related to entering a community a community of peers and thus allows for social exchange and/or competition.
- Provision of meaning: Gamification frequently allows users to participate in resolving superordinate problems that go beyond one's personal possibilities [3].

These mechanisms facilitate perceptions of control, autonomy, and fun that are central antecedents of flow experiences. Flow reinforces the voluntary use of gamified service bundles and increases both motivation and performance of individuals [3]. However, intrinsic motives and flow can be systematically activated by setting extrinsic incentives [9]. Thus, incentives such as badges do not only com- ply with the intrinsic motive of collecting but also with the extrinsic motive of gaining social recognition. As a consequence, gamification allows for the de- sign of persuasive incentive mechanisms that go far beyond financial incentives. Thus, gamification has a high potential for changing behavioural patterns and for supporting accompanying learning processes:

- Behavioural change: Gamification relates behavioural change to positive emotional feedback. In doing so, gamification may support the introduction of new patterns of behaviour as well as the modification of habitual behaviour. Such behavioural patterns are usually unconscious and automatized so that traditional incentive schemes frequently only exhibit a low effective- ness. In providing positive emotions, gamification may break up existing habits, update them with new behaviours and support the stabilization of new behaviours by continuously setting appropriate stimuli [10].
- Support of learning processes: Gamification decomposes tasks into various subtasks and milestones. Users can solve such subtasks by trial and error and repeat them until the problem has been solved and a particular skill level has been reached [3]. By designing tasks of increasing difficulty, cognitive structures for the internalization of learning contents may be systematically created [11].

2-2- The MDA Framework

Developing a game is often difficult as it has many facets and requires technical skills in various areas, such as illustration, sound and storytelling, but also base mechanisms of game systems, the overarching design goals, or desired experiential results of gameplay. Developing a game is often done by people from different disciplines in the fields from creativity to scientific backgrounds who eventually congregate. In an attempt to bridge the gap between game design and development, game criticism, and technical game research, the MDA framework was developed. The

MDA framework is the most frequently used framework for Game Design and Game Research and was presented by Hunicke et al. It was developed and thought as part of the Game Design and Tuning Workshop at the Game Developers Conference, San Jose 2001-2004 and is uzed to analyze games. The MDA framework standing for Mechanics, Dynamics and Aesthetics is a formal approach for understanding games. The three levels of the MDA framework (Figure.1) make it possible to conceptualize the dynamic behavior of game systems. Games are dynamic systems and understanding them will help to develop techniques for resumable design and improvement. It will help tune for desired behavior and control undesired outcome. 'The MDA framework allows us to reason explicitly about particular design goals, and anticipates how changes will impact each aspect of the framework and the resulting design/implementations [12].

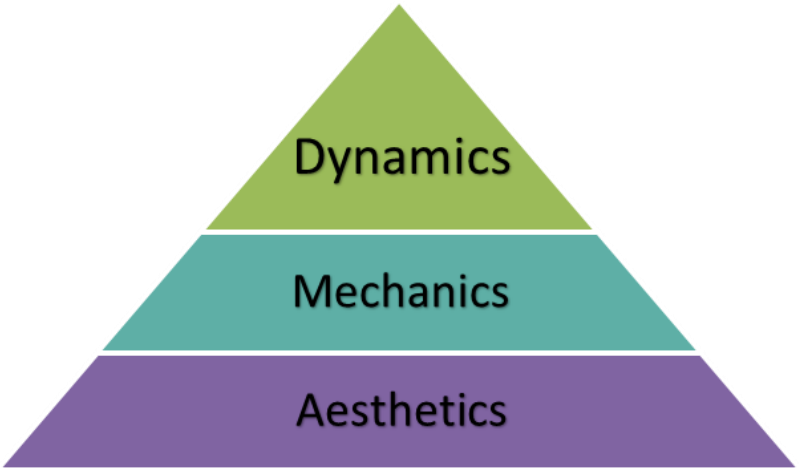

Figure 1. MDA framework

The mechanics refer to the functioning components of the game; they allow the designer to come up with the levers of the game, which will guide the player to the desirable behavior. It describes certain game elements that encourage players to do a specific action. Think, for example, rules, goals, points, badges, levels virtual goods, virtual environments and scoreboards. The Dynamics determine what each player is doing in response to the mechanics of the system and therefore interact with the player. Dynamics can only be provoked and cannot be directly influenced. Therefore it is also important to test if the desired behavior is generated with the mechanics at an early stage. At last the Aesthetics, which is the most difficult and most important part, is responsible for making the game fun. The aesthetics set the tone for the player. It is about how the player feels during the action [13].

2-3- The Proposed Model of Motivation in Gamification

Motivation to act has been studied in Social Psychology, Educational Psychology, and Organizational Science. These areas focus on motivation in particular types of environments. We propose to link gamifi cation to these theories. Motivation is demonstrated by an individual's choice to engage in an activity and the intensity of effort or persistence in that activity. Current approaches concern two dominant clusters that play a role in determining player's motivation: extrinsic and intrinsic motivation [9, 14].Gamification combines these two motivations; on one hand using extrinsic rewards such as levels, points, badges to improve engagement while striving to raise feelings of achieving mastery, autonomy, sense of belonging [15].

Notably, the social aspect is important in games [16]. Competition, social interaction, or cooperation may infl uence player behavior [17-19]. Hence, following Vassileva [20], the present approach covers a spectrum of motivations from extrinsic, through social, to intrinsic (Figure 2). At one extreme of the spectrum, we place extrinsic motivation which is the focus of Expectancy Value Theory and Skinner's Reinforcement Theory. These theories explain the motivation to perform actions or behaviors that induce extrinsic rewards [20]. On the other end of the spectrum, intrinsic motivations are the focus of Maslow's Hierarchy of Needs, Atkinson's Need Achievement Theory, as well as Bandura's Self-Effi cacy Theory and Goal Setting Theory. All these are need-based theories. Theories in the middle of the spectrum explain the social motivation of games. In this context we identify Festinger's Social Comparison and Personal Investment Theory (PIT). Specifi c references for each theory are given in the following sections. Additionally, we consider Deci and Ryan's Self-Determination Theory as a comprehensive theory since it encompasses both intrinsic and extrinsic motivations on a continuum from internal to external motivation [21]. The following brief overview provides the highlights of each theory togetherwith its specifi c application in games.

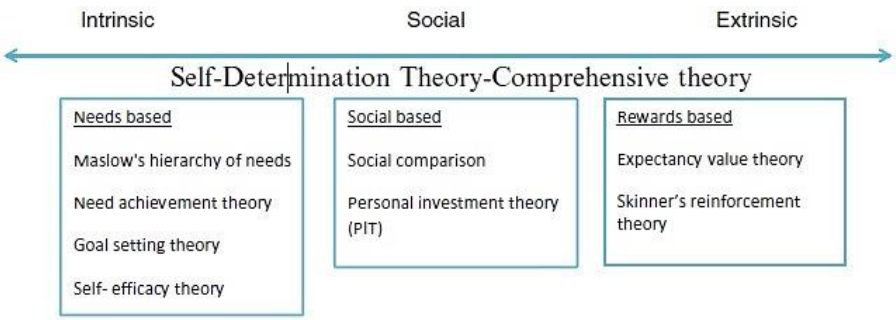

Figure 2. Model of motivation in games [20, 21]

3- Needs-Based Theories

Needs based theories relevant for studying intrinsic motivations in serious games include Maslow's hierarchy of needs, Need Achievement Theory, Goal Setting and Self Efficacy which are briefly described in the following.

3-1- Hierarchies of Needs

One of the earliest and best known theories of motivation comes from the psychologist Abraham Maslow. According to Maslow, human behaviors are driven by the desire to satisfy physical and psychological needs. Maslow proposes five levels of needs that drive human activities, ranging from physiological needs to the need for self-actualization [22]. According to the hierarchy of needs we must satisfy physiological needs and needs for safety and security before progressing to more complex needs such as desire for belongingness, self-esteem and finally self-actualization (Figure .3a). As we progress up Maslow's hierarchy we move away from needs that are produced by deficiencies to needs produced by positive goals and incentives [22, 23]. Based on Maslow's hierarchy of needs, Siang et al. [24] illustrate game players' needs where the lower levels need to be fulfilled before any of the higher levels in the pyramid (Figure. 3b). At the bottom level, players seek information to understand the basic rules of game. Once the rules need is satisfied, players need safety, information for persisting and winning. The third level refers to belongingness need in which players need to feel comfortable with the game and eventually achieve the game goal. After knowing that winning is possible, there is a need to feel good when playing the game—a feeling of esteem. At the next level, players start to expect a greater challenge, they need to understand and know more about the game such as different strategies. The sixth level is an aesthetic need which reflects the call for good graphics, visual effects, appropriate music, sound effects, etc. Finally, players want to be able to do anything within the game rules and constraints (attaining a form of perfection in the virtual world) [25].

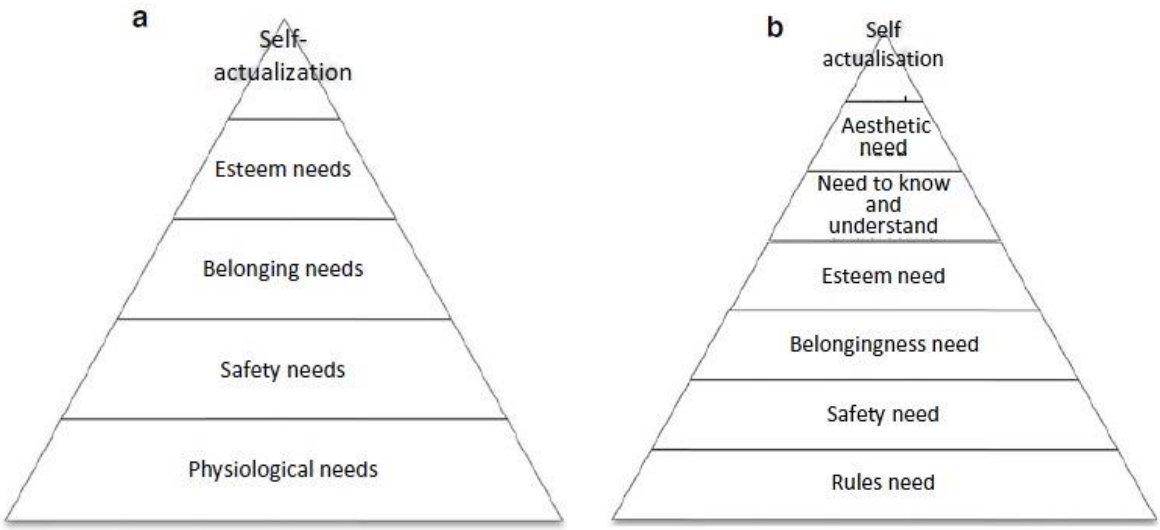

Figure 3. (a) Maslow's hierarchy of needs, (b) Hierarchy of players' needs licensed from [23, 24]

3-2- Need Achievement Theory

Achievement behaviour is directed at developing or demonstrating, to self or to others, high rather than low ability [26, 27]. It implies that in achievement situations people desire success to the extent that it indicates high ability and

seek to avoid failure which may be a signal of low ability [27]. According to Atkinson, to achieve success and to avoid failure are two separate motives. These two motives affect the level of task difficulty people choose to undertake. People with high motivation to succeed prefer tasks of intermediate difficulty [26]. Games often display achievement systems and status indicators. These systems aim to encourage game play and to monitor performance [28]. According to Montola, et al [29], achievements systems are reward structures providing additional goals for players and hence they trigger some friendly competition and comparison among users [29].

3-3- Goal Setting Theory

A goal is what the individual is trying to accomplish, the object or aim of an action [30]. Goal setting theory claims that difficult, specific, context appropriate, and immediate goals, rather than long-term goals, motivate to achieve more [16]. Goals affect performance by directing attention, assembling effort, increasing persistence and belief in ability to complete a task [30]. Goal setting is most likely to improve task performance when the goals are specific and sufficiently challenging, the subjects have sufficient ability, feedback is provided to show progress in relation to the goal, rewards are given for goal attainment, and the assigned goals are actually accepted by the individual [30, 31].

3-4- Self-Efficacy

Self-efficacy refers to perceived performance ability for a specific activity [32]. Judgment of self-efficacy determines choice of activities, select challenging settings, effort expended, persistence and task performance [32, 33]. Self-efficacy levels can enhance or impede motivation. People with high self-efficacy choose to perform more challenging tasks. They invest more effort; they persist; and when failure occurs they recover more quickly and maintain the commitment to their goals [33].

4- Social-Based Theories

Social Comparison Theory and Personal Investment Theory elaborate the social side of games. Social comparison states that people seek to evaluate their beliefs, attitudes and abilities by comparing their reaction with others. Personal investment theory suggest that the level to which a person will invest personal resources of effort and time for an activity depends on personal incentives, beliefs regarding oneself, and comprehended alternatives [34].

4-1- The Social Comparison Theory

An important source of knowledge about oneself is comparisons with other people. This insight is the basis of The Social Comparison Theory presented by Festinger [35, 36]. According to the theory we evaluate our beliefs, abilities, and reactions by comparing them with those of others [37, 22]. Festinger's "similarity hypothesis" predicts that people compare themselves with similar others [35, 37].

4-2- Personal Investment Theory (PIT)

Personal Investment Theory (PIT) integrates social influences with the examination of achievement motivation [38]. PIT holds that the meaning a person creates in the form of beliefs, perceptions, feelings, purposes, and goals motivates behaviour. These cognitive elements are the key to understand and predict investment behaviour such as participation, spending of time and effort. Specifically, the theory defines three basic components of meaning as critical to determining personal investment in specific situations: personal incentives, sense of self, and perceived options [38, 39].

Games use incentives as motivational hooks that maintain interest and help to stretch engagement and repeat usage. Generally, incentives are reliant on some aspect of performance [40].

5- Rewards-Based Theories

On the right side of the spectrum (Figure 2) extrinsic motivations are created through external factors, rewards, or incentives [41]. We refer to two main theories in this regard: Expectancy Value Theory and Skinner Reinforcement Theory.

5-1- Expectancy Value Theory (EVT)

Expectancy value theory relates to the strength of motivation to strive for a certain goal, to the expectations to attain the desired goal, and to the incentive value of that particular goal [42]. Expectancy value theory holds that goal directed behaviour is a function of the belief that efforts will lead to performance needed to attain the rewards; performance will determine the outcome; and the value attached to achieving the outcome [43]. The theory argues that expectancies and values influence achievement choices, persistence, effort, and performance [44-46].

5-2- Skinner's Principle of Partial Reinforcement

Reinforcement as understood by Skinner constitutes outcomes that strengthen the probability of a response [22, 47]. Skinner noted that continuous reinforcement establishes desired behaviours quicker than partial reinforcement. But once the continuous reinforcement is removed, the desired behaviours extinguish quickly. According to his principle of partial reinforcement occasional reinforcement of behaviours leads to a greater persistence to extinction than continuous reinforcement [22]. Variable ratio schedules are more effective than fixed ratio in sustaining desired behaviours. Gambling and lottery games are good examples of a reward based on a variable ratio schedule [48].

5-3- Self-Determination Theory

Self-Determination Theory (SDT) focuses on types, rather than amount, of motivation, paying particular attention to autonomous motivation, controlled motivation, and a motivation as predictors of performance and well-being (Figure 4).

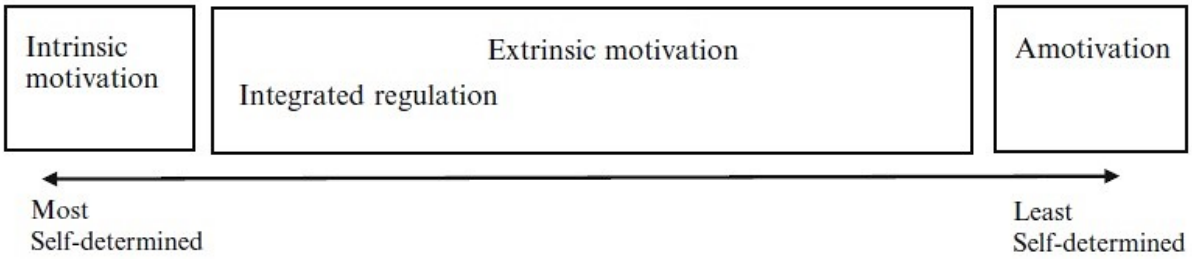

Figure 4. The concept of a continuum from internal to external motivation [9]

SDT proposes that motivation is multidimensional and resides along a continuum of self-determination ranging from intrinsic motivation through extrinsic motivation to a motivation [21, 49]. SDT discusses three psychological needs: autonomy, competence, and relatedness [9, 50, 51]. Autonomy is the ownership of one's behaviour. Competence is the ability to produce desired outcomes and to experience mastery and effectiveness. Relatedness is the feeling of being connected with others. If these three needs are satisfied, growth and development results, and intrinsic motivation for the task increases. When the three needs are not met, negative emotions (anxiety and anger) may result, and intrinsic motivation is undermined [51]. Studies of SDT and education have shown that supporting intrinsic needs of autonomy, competence, and relatedness facilitates deeper and more internalized learning. Recent studies confirmed that experiences of competence, autonomy, and relatedness were major contributors to game enjoyment, regardless of the specific content, complexity, or genre of games [51, 52].

6- Conclusion

This chapter's focus is on contributing to the body of knowledge as it relates to the measurement of gamification outcomes, i.e. a methodology for the assessment of gamification implementation within organizations. The term gamification has come to the forefront with much fanfare and receptivity by both educators and corporate training professionals. One only has to look online to see the myriad number of conferences, publications and blogs devoted to the topic and perpetuation of its perceived benefits. (Figure.5), below indicated the number of titles on the topic illustrating continuous increases in the subject matter over the last four years. Yet despite the large amount of hits on the topic, there still remains a lack of coherent understanding on what kinds of studies and results gamification has yielded. Moreover, understanding the effectiveness of gamification remains a pertinent issue.

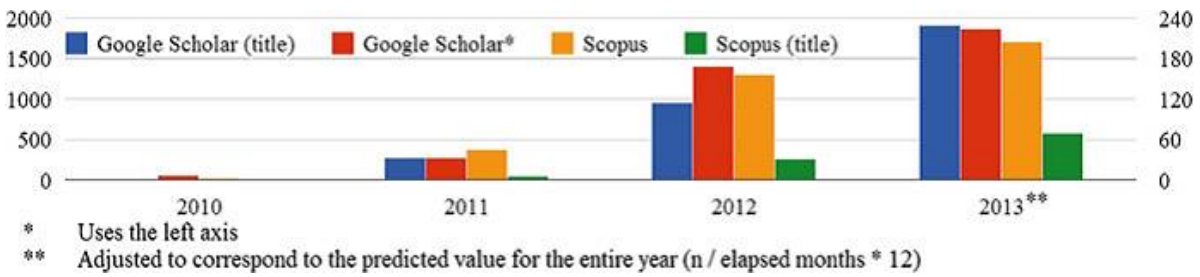

Figure 5. Search hits for "gamification"

The term gamification as defined by Huotari and Hamari is as follows:

- A process of enhancing a service with (motivational) affordance in order to invoke gameful Experiences and further behavioural outcomes.

As such the use of gamification involves the utilization of several of the concepts inherent in games such as level design, tokens, badges and other rewards to incentivize learning during play. What makes the use of gamification unique when pitted against traditional learning and development methodologies both in education and business is the element of play?

Gamification is a secondary phenomenon based on a game as its origin. The game has been studied in details from the standpoint of the various sciences-psychology, sociology and cultural studies. What kind of immanent characteristics of the game did give impetus for the modern development of the phenomenon? The process of the game has some special characteristics, which can be divided into several categories. The first one includes voluntary participation, freedom from pressure and utilitarian purposes, the ability to pass through experience of strong emotions and catharsis, to risk and win. Secondly, there is an existence in an illusory world, a temporary escape from the reality, building of a gaming space with its special rules and the repeatability of scenarios in a number of options. Finally, the last category of game characteristics includes lack of external motivation, value of the game process and self-sufficiency. The foundation of Gamification is the game mechanics. Therefore In terms of mental and psychological function, Gamification works quite similar game. Therefore, knowledge of Gamification performance in term of mental and psychological is very important as a prerequisite for those who want to design and implement Gamification projects.

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